Practical No. 13

Data-Driven Testing Using Excel Integration

Aim: Write a program to update 10 student records from an Excel file and validate the changes in a web application using Apache POI.

Steps

1. Create folder **D:\DataDrivenTesting**.
2. Create **StudentData.xlsx** with a sheet named **Students**; use columns Student ID, Name, Course, and Marks, with 10 student records. Save it in the folder.
3. Create **StudentForm.html** in the same folder and paste the HTML code below.
4. Create a Maven project named **DataDrivenTesting** and add the Selenium and Apache POI dependencies in **pom.xml**.
5. Create package **com.selenium.practical** and class **DataDrivenTest**.
6. Paste the Java code, save, and run as **Java Application**.

Code

StudentForm.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Management</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 40px;
            background-color: #f4f4f4;
        }
        .container {
            width: 500px;
            background-color: white;
            padding: 25px;
            border: 1px solid #ccc;
            border-radius: 8px;
        }
        h2 {
            color: #333;
        }
        label {
            display: inline-block;
            width: 120px;
            margin-bottom: 15px;
        }
        input {
            width: 250px;
            padding: 8px;
            margin-bottom: 10px;
        }
        button {
            padding: 10px 20px;
            background-color: #007bff;
            color: white;
            border: none;
            cursor: pointer;
            border-radius: 4px;
        }
        button:hover {
            background-color: #0056b3;
        }
        #result {
            margin-top: 20px;
            padding: 15px;
            color: green;
            font-weight: bold;
            border: 1px solid #ccc;
        }

    </style>
</head>
<body>
    <div class="container">
        <h2>Student Management System</h2>
        <form id="studentForm">
            <label>Student ID:</label>
            <input
```

```
                type="text"
                id="studentId">
            <br>
            <label>Student Name:</label>
            <input
                type="text"
                id="studentName">
            <br>
            <label>Course:</label>
            <input
                type="text"
                id="course">
            <br>
            <label>Marks:</label>
            <input
                type="text"
                id="marks">
            <br><br>
            <button
                type="button"
                id="updateButton"
                onclick="updateStudent()">
                Update Student
            </button>
        </form>
        <p id="result"></p>
    </div>
    <script>
        function updateStudent() {
            var id =
                document.getElementById("studentId").value;
            var name =
                document.getElementById("studentName").value;
            var course =
                document.getElementById("course").value;
            var marks =
                document.getElementById("marks").value;
            document.getElementById("result").innerHTML =
                "Student Updated: "
                + id
                + " | "
                + name

                + " | "
                + course
                + " | "
                + marks;
        }
    </script>
</body>
</html>
```

pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.selenium</groupId>
    <artifactId>DataDrivenTesting</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <properties>
        <project.build.sourceEncoding>
            UTF-8
        </project.build.sourceEncoding>
        <maven.compiler.source>21</maven.compiler.source>
        <maven.compiler.target>21</maven.compiler.target>
    </properties>
    <dependencies>
        <!-- Selenium WebDriver -->
        <dependency>
            <groupId>
                org.seleniumhq.selenium
            </groupId>
            <artifactId>
                selenium-java
            </artifactId>
            <version>4.35.0</version>
        </dependency>
        <!-- Apache POI -->

        <dependency>
            <groupId>
                org.apache.poi
            </groupId>
            <artifactId>
                poi-ooxml
            </artifactId>
            <version>5.4.1</version>
        </dependency>
```

```
    </dependencies>
</project>
```

DataDrivenTest.java

```
package com.selenium.practical;
import java.io.File;
import java.io.FileInputStream;
import org.apache.poi.ss.usermodel.DataFormatter;
import org.apache.poi.ss.usermodel.Row;
import org.apache.poi.ss.usermodel.Sheet;
import org.apache.poi.ss.usermodel.Workbook;
import org.apache.poi.ss.usermodel.WorkbookFactory;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;
public class DataDrivenTest {
    public static void main(String[] args) {
        WebDriver driver = null;
        FileInputStream file = null;
        Workbook workbook = null;
        try {
            // Excel file path
            String excelPath = "D:\\DataDrivenTesting\\StudentData.xlsx";
            // HTML file path
            String htmlPath = "file:///D:/DataDrivenTesting/StudentForm.html";
            // Check Excel file
            File excelFile = new File(excelPath);

            if (!excelFile.exists()) {
                throw new RuntimeException("Excel file not found: " + excelPath);
            }
            // Open Excel file
            file = new FileInputStream(excelFile);
            System.out.println("Excel file opened successfully.");
            // Create workbook
            workbook = WorkbookFactory.create(file);
            // Select Students sheet
            Sheet sheet = workbook.getSheet("Students");
            if (sheet == null) {
                throw new RuntimeException("Students sheet not found.");
            }
            System.out.println("Students sheet selected successfully.");
            // Create data formatter
            DataFormatter formatter = new DataFormatter();
            // Launch Chrome
            driver = new ChromeDriver();
            driver.manage().window().maximize();
            System.out.println("Chrome launched successfully.");
            // Set implicit wait
            driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
            // Create explicit wait
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
            // Open student application
            driver.get(htmlPath);
            System.out.println("Student Management System opened.");
            // Process 10 student records
            for (int i = 1; i <= 10; i++) {
                // Get Excel row
                Row row = sheet.getRow(i);
                if (row == null) {
                    System.out.println("Row " + i + " is empty. Skipping.");
                    continue;
                }
                // Read student ID
                String studentId = formatter.formatCellValue(row.getCell(0));
                // Read student name
                String name = formatter.formatCellValue(row.getCell(1));
                // Read course
                String course = formatter.formatCellValue(row.getCell(2));
                // Read marks
                String marks = formatter.formatCellValue(row.getCell(3));
                System.out.println();
                System.out.println("Processing Student: " + studentId);
                // Enter student ID
                WebElement studentIdField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("studentId")));
                studentIdField.clear();
                studentIdField.sendKeys(studentId);

                // Enter student name
                WebElement studentNameField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("studentName")));
                studentNameField.clear();
                studentNameField.sendKeys(name);
                // Enter course
                WebElement courseField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("course")));
```

```
                courseField.clear();
                courseField.sendKeys(course);
                // Enter marks
                WebElement marksField =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("marks")));
                marksField.clear();
                marksField.sendKeys(marks);
                // Click Update Student button
                WebElement updateButton =
wait.until(ExpectedConditions.elementToBeClickable(By.id("updateButton")));
                updateButton.click();
                // Read result
                WebElement resultElement =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("result")));
                String result = resultElement.getText();
                System.out.println("Web Result: " + result);
                // Verify student data
                if (result.contains(studentId) && result.contains(name) && result.contains(course)
&& result.contains(marks)) {
                    System.out.println("PASS - Student " + studentId + " data verified successfully.");
                } else {
                    System.out.println("FAIL - Student " + studentId + " data verification failed.");
                }
                // Pause before next record
                Thread.sleep(500);
            }
            // All records processed
            System.out.println();
            System.out.println("All 10 student records processed.");
        } catch (Exception e) {
            // Handle exception
            System.out.println();
            System.out.println("An error occurred:");
            e.printStackTrace();
        } finally {
            // Close browser
            if (driver != null) {
                driver.quit();
                System.out.println("Browser closed successfully.");
            }
            // Close workbook

            if (workbook != null) {
                try {
                    workbook.close();
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
            // Close Excel file
            if (file != null) {
                try {
                    file.close();
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }
        }
    }
}
```